

Advances in polymers for anti-biofouling surfaces

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Self-assembling polymers and nanostructured polymer thin films are being actively explored as advanced coatings for marine and biomedical applications. This review highlights recent advances in the design and synthesis of polymers that can resist fouling by biomolecules, cells and organisms. Current understanding of the mechanisms of anti-biofouling activity is also discussed.

Introduction

Anti-biofouling surfaces lie at the heart of several contemporary and advanced technologies, ranging from coatings for ship hulls¹ and biomedical implants,² to biosensors³ and carriers for targeted drug delivery.⁴ In general, antifouling surfaces are based on minimizing the intermolecular forces of interactions between extracellular biomolecules and the synthetic surface so that an adhered cell is easily released under low shear stresses. A truly antifouling surface, also referred to as a 'stealth surface', remains undetected by cells in a biological milieu, and is therefore not prone to biofouling. Prior research on cell-material interactions has identified two main classes of non-toxic coatings that can resist biofouling. Hydrophobic surfaces such as those of poly(dimethylsiloxane) (PDMS) are

known to show good 'fouling release' properties, whereas the hydrophilic surfaces of polymers like poly(ethylene glycol) (PEG), with low values of polymer-water interfacial energy, show resistance to protein adsorption and cell adhesion. The interfacial energy between a surface and water is expected to play an important role in conferring antifouling characteristics to a surface.⁵ Non-polar hydrophobic surfaces exhibit high interfacial energy with water. For example, the interfacial energy of PDMS with water is about 52 mJ m⁻².⁶ In comparison, the water interfacial energy of polar polymers such as PEG or poly(2-hydroxyethyl methacrylate) is below 5 mJ m⁻².⁷ Consequently, when a hydrophobic surface contacts a biological interface, amphiphilic biomolecules such as proteins show significant adsorption on the surface to minimize the interfacial energy. Nevertheless, non-polar surfaces of elastomeric polymers such as PDMS have been widely explored as fouling release coatings, because their low modulus and low surface energy allow the adhered organism, such as a barnacle, to easily detach from the surface. The

adsorption of biomolecules on hydrophilic surfaces, however, does not confer any significant thermodynamic advantage because the interfacial energy is already sufficiently low. These surfaces, therefore, show resistance to protein adsorption.

Fig. 1 depicts the general concept of an anti-biofouling surface using the example of zoospores of marine green alga *Ulva*. *Ulva* is a predominant fouling species in marine biofilms. Motile zoospores of *Ulva* swim using whip-like whiskers called flagella, actively seeking surfaces for settlement. Upon 'detection' of a suitable surface, the spore settles, attaches to the substrate by its anterior flagellated end, and both absorbs the flagella into the protoplasm and secretes a glycoprotein adhesive present in membrane bound cytoplasmic vesicles (Fig. 1a).⁸ After settlement and attachment, spore germination occurs giving rise to sporelings, which further adhere to the substratum using adhesives secreted by the rhizoids. An anti-biofouling coating can prevent detection of the surface by the zoospores and evades settlement and attachment (Fig. 1b). A fouling release coating, on the other hand, enables easy release of the

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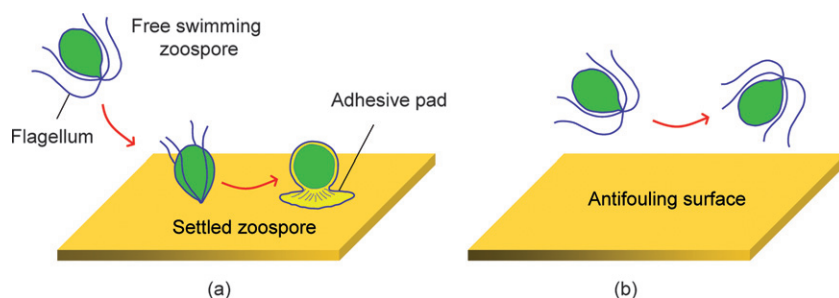


Fig. 1 (a) Settlement and adhesion of a zoospore of the green alga *Ulva* on a surface prone to biofouling, and (b) spore behavior near an antifouling surface (ref. 8).

adhesive pad under moderate shear stresses generated by flow of water past the surface, so that the zoospores (or the sporelings) cannot remain attached to the substrate.

In recent years, with advances in macromolecular synthesis and self-assembly, several polymer coatings have been designed and explored as anti-biofouling surfaces. The performance of these coatings is based on the aforesaid principles, namely, low interfacial energy with water, low intermolecular forces of interaction with biomolecules, or both. The purpose of this review is to highlight novel polymer architectures and surface modification strategies that impart anti-fouling characteristics to a surface. Fig. 2 summarizes the three major non-toxic approaches to make a surface resistant to biofouling. Much of this work is part of two major research efforts, one funded by the US Office of Naval Research (<http://www.onr.navy.mil>) and the other by the European Community through its AMBIO program (<http://www.ambio.bham.ac.uk/>). Both programs are actively studying new concepts in environmentally friendly fouling prevention, which are desperately required to replace traditional inherently toxic metal-based systems of fouling control.

The following highlight will summarize different approaches taken by various research groups to combat the challenge of forming anti-biofouling and fouling release surfaces. Particular attention will be given to the chemical structures produced and the various techniques utilized to demonstrate these surfaces' inherent anti-biofouling character. The performance evaluation of these anti-fouling polymers has been greatly facilitated by marine biologists including the research groups of Professor James

Callow (University of Birmingham) which study settlement and adhesion of marine algae, and Professors Dean Wendt (California Polytechnic Institute), Geoffrey Swain (Florida Institute of Technology), and Anthony Clare (New Castle University) who conduct laboratory and

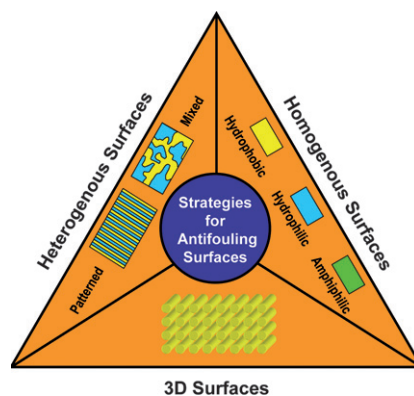


Fig. 2 Three major approaches to producing anti-biofouling surfaces. Homogeneous surfaces of hydrophilic (blue), hydrophobic (yellow), or amphiphilic (green) character can be utilized. Protein and cell resistant hydrophilic surfaces help to avoid the settlement of fouling organisms in the first place. The inherent 'slipperiness' of non-polar low surface energy, hydrophobic coatings leads to facile release. In combination with low surface modulus, these coatings can aid the detachment of settled organisms under applied shear flow. Amphiphilic surfaces with chemical or textural complexity can effectively 'confuse' fouling organisms into avoiding settlement and are effective against a vast range of biomolecules, cells, and organisms in a biological environment. Mixed or patterned surfaces of alternating hydrophobic and hydrophilic character can also be utilized due to their intricacy. Finally, 3D surfaces can be used to modify the surface of an anti-biofouling coating by control of surface topography. This factor can influence the ability of marine organisms to stick to the coating surface.

field assays involving barnacles and other marine organisms. While the focus of the highlight will be on the role of materials chemistry in anti-biofouling, there are other promising antifouling approaches that rely solely on modifying physical characteristics of the coating. A prominent outcome of the latter approach is the shark-skin inspired PDMS coating, with the Sharklet AF™ microtopography, developed by the research group of Professor Anthony Brennan (University of Florida).⁹ Such 3D coatings (Fig. 2) based on microtopography manipulations will not be further discussed in this article.

Liquid crystalline block copolymers with semifluorinated side chains

Self-assembly of block copolymers with liquid crystalline semifluorinated alkyl side chains can be used to create a highly non-polar hydrophobic surface. The block copolymer architecture shown in Fig. 3 consists of a polystyrene block that imparts solubility to the fluorinated polymer in common organic solvents, and also acts as a compatibilizer when the surface-active block copolymer is blended with a relatively inexpensive thermoplastic elastomer in the coating formulation.¹⁰

The fluorinated block, with mesogenic semifluorinated alkyl side chains, preferentially segregates to the air-polymer

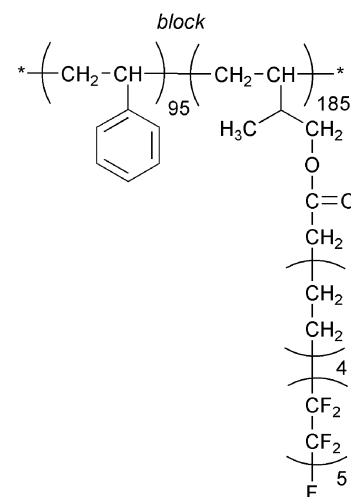


Fig. 3 Block copolymer with liquid crystalline semifluorinated alkyl side chains that form a self-assembled monolayer of non-polar fluoroalkyl groups at the surface of the coating.

Block copolymers with environmentally responsive amphiphilic side chains

$$\begin{array}{c} \text{block} \\ \left(\text{CH}_2 - \underset{\text{C}_6\text{H}_5}{\text{CH}} \right)_{82} - \left(\text{CH}_2 - \underset{\begin{array}{c} \text{C}=\text{O} \\ | \\ \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{CF}_2 \\ | \\ \text{CF}_2 \\ | \\ \text{F} \end{array}}{\text{CH}} \right)_{23} \end{array}$$

surface wettability. These surfaces, that combined the non-adhesive properties of non-polar fluoroalkyl groups with the antifouling characteristics of hydrophilic PEGylated groups, resulted in weak adhesion of both *Ulva* and *Navicula*,¹³ low settlement density of barnacle larvae,¹⁴ and excellent resistance to protein adsorption. Tapping mode scanning probe microscopy indicated that the amphiphilic block copolymer surfaces used in the marine biofouling assays were topographically smooth and compositionally uniform, with a lamellar morphology parallel to the substrate. The

Perfluoropolyether-based random terpolymers

PEGylated polymers

$$\begin{array}{c} \text{ran} \qquad \qquad \qquad \text{ran} \\ \left(\text{CH}_2 - \underset{\begin{array}{c} \text{C=O} \\ | \\ \text{OCH}_3 \end{array}}{\overset{\text{CH}_3}{\text{C}}} \right)_{0.3} \left(\text{CH}_2 - \underset{\begin{array}{c} \text{C=O} \\ | \\ \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{CH} \\ / \quad \backslash \\ \text{O} \quad \text{CH}_2 \end{array}}{\overset{\text{CH}_3}{\text{C}}} \right)_{0.3} \left(\text{CH}_2 - \underset{\begin{array}{c} \text{C=O} \\ | \\ \text{NH} \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{NH} \\ | \\ \text{C=O} \\ | \\ \text{F}_3\text{C}-\text{CF} \\ | \\ \text{O} \\ | \\ \text{CF}_2 \\ | \\ \text{F}_3\text{C}-\text{CF} \\ | \\ \text{F} \end{array}}{\overset{\text{CH}_3}{\text{C}}} \right)_{0.4} \\ \sim 6 \end{array}$$

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Navicula diatoms compared to PDMS. Diatoms are known to adhere strongly to hydrophobic surfaces, which makes commercial PDMS based coatings susceptible to accumulation of diatom slimes. Therefore, developing coatings that can resist diatom fouling is an important area of marine biofouling research. A recent advance in hydrophilic PEGylated coatings is the use of bio-inspired polymers prepared from methoxy-terminated PEG and the adhesive amino acid L-3,4-dihydroxyphenylalanine (DOPA).¹⁷ In this approach, mimics of mussel adhesive proteins are conjugated with antifouling polymers to confer antifouling properties to surfaces. The catechol side groups of DOPA are capable of forming moisture resistant bonds with a variety of substrates including metals, glass, and plastics.¹⁸ Titanium surfaces coated with the polymer shown in Fig. 6 were found to have good antifouling and fouling release properties in marine biofouling assays using both green algal spores and diatoms.¹⁷ These surfaces also showed good resistance to protein and mammalian cell fouling.¹⁹ Other approaches for creating PEGylated antifouling surfaces include the use of dendrimers and hyperbranched polymers. Zhao *et al.* found that membranes prepared from blends of hydrophobic poly(vinylidene fluoride) (PVDF) and a hyperbranched polymer with hydrophilic PEG grafts showed significantly lower protein adsorption than pure PVDF membranes.²⁰

As long as a surface is completely covered by EG groups, the PEG chain length and the entropic effects associated with the molecular architecture do not have a strong influence on antifouling activity. Complete surface coverage by PEG can be achieved if the distance between the attachment sites of the PEG

grafts is less than the radius of gyration of PEG in solution.²¹ In other words, the grafting density must be in the polymer brush regime, a condition that is readily achieved using comblike polymers with high grafting densities along the polymer backbone.¹⁰ The principal advantage of longer PEG grafts, comblike polymers with PEGylated side chains, PEGylated polymer brushes, or dendritic architectures is the formation of a dense, defect-free coating of EG groups at the surface.

Self-assembled monolayers with only a few EG units per molecule have shown remarkable resistance to adsorption of a variety of proteins.²² Experiments with such SAMs give credence to the hypothesis that the formation of a hydration layer near a hydrophilic surface is a more general basis for protein resistance than the steric repulsion mechanism that is usually invoked in the case of longer PEG chains. Heuberger *et al.* have found that the water content inside surface-grafted PEG chains is over 80 vol%, and have proposed that the excellent protein resistance properties of this type of surface were a result of the high degree of organization in the PEG–water complex.²³ Protein adsorption requires disruption of this complex, which is energetically and kinetically unfavorable.

A counterargument is the report by Jiang *et al.*, who produced hyperbranched PEG-like structures (Fig. 7) on the surfaces of silicone elastomers and compared their antifouling properties with linear PEG grafts.²⁴ Both of these surfaces are expected to be highly hydrated due to densely packed hydroxyl and PEG groups. The researchers found that linear PEG molecules were much more effective in preventing bacterial attachment than branched architectures, and attributed this effect to the differences in the magnitudes of steric repulsion

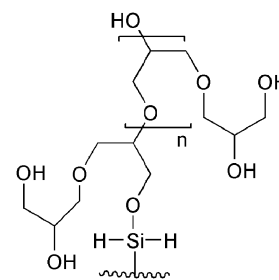


Fig. 7 PEG-like hyperbranched polymer with antifouling properties. The hydroxyl groups were also further reacted with PEG molecules to obtain hyperbranched PEGylated surfaces.

between the linear and branched molecular arrangements. They have, however, not reported a comparison of the protein resistance of linear and branched PEG architectures. Protein adsorption assays involving a direct comparison of linear and branched surface structures can resolve the ongoing debate on the hydration *versus* steric repulsion mechanisms for antifouling activity.

Benhabbour *et al.* have recently reported a similar study where a thiol-terminated poly(ethylene glycol) (HS–PEG₆₅₀–OH) was attached to gold-coated silicon wafers.²⁵ The terminal PEG OH groups were functionalized with aliphatic polyester dendrons of generations 1 to 4. They found that although dendronization of the PEGylated surfaces resulted in an increase in surface hydrophilicity, protein adsorption increased. Benhabbour *et al.* have proposed that PEG chain flexibility is one of the key factors in the mechanism of protein resistance, and that the chain flexibility was impeded by introduction of dendrons with multiple peripheral OH groups. The interaction of the OH groups with the underlying PEG is believed to lower the conformational flexibility of the PEG grafts.

PEGylated polymers are currently among the best protein resistant surfaces. Their long-term stability in a biological environment is crucial in practical applications as antifouling coatings. Sharma *et al.* found that the PEG-modified silicon surfaces retained their protein and cell repulsive properties even after at least 4 weeks of immersion in a PBS buffer (pH 7.4) at 37 °C, in 5% CO₂.²⁶ Similar studies that investigate the durability of PEG coatings in a marine environment are necessary. Polyethers and many

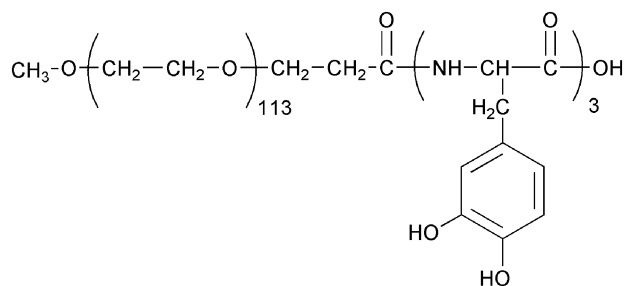


Fig. 6 Methoxy-terminated PEG, conjugated to DOPA tripeptide to promote adhesion to metal, glass and polymer substrates.

other hydrophilic polymers can undergo oxidative degradation and chain cleavage.²⁷ Alternatives to PEG-based antifouling coatings are being actively researched.

Hyperbranched amphiphilic fluoropolymers

Based on the hypothesis that nanoscale heterogeneities in topography and composition will create a surface that is unfavorable for protein adsorption, Gudipati *et al.* have investigated surfaces of hyperbranched polymers containing fluorinated and PEGylated groups.²⁸ At an optimal composition of hydrophobic and hydrophilic monomers in the polymer, low protein adsorption and high fouling release was achieved. More recently, Powell *et al.* have reported the synthesis of hyperbranched fluoropolymers using atom transfer radical polymerization of an inimer shown in Fig. 8.²⁹ They proposed that a coating prepared using the resulting polymer would have surface heterogeneities small enough in size to enhance resistance to protein adsorption and cell adhesion. While a textured surface is expected to provide a greater surface area for interaction with proteins (and is detrimental for antifouling activity), the design principle of hyperbranched amphiphilic polymer coatings is that the nonuniformity of surface characteristics would adversely affect the ability of a protein molecule to adsorb and unfold on the surface. Indeed, for this premise to be valid, the size of the heterogeneities must be below the size of the protein molecules, in the 1–10 nm range. Definitive evidence for the fact that a nanoscale variation in surface topography and chemistry can impart superior resistance to protein adsorption (than a topographically

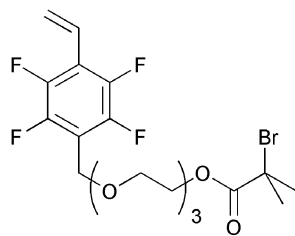


Fig. 8 Amphiphilic monomer with an initiating group for atom transfer radical polymerization.

smooth and chemically uniform surface) is still lacking. Nevertheless, recent studies have shown that marine organisms can discriminate between surface domains of different wettability and of a specific size in the micrometre range.³⁰

Patterned fluorinated and PEGylated monolayer surfaces

To facilitate the design of 'ambiguous' surfaces made to present both hydrophobic and hydrophilic domains to fouling organisms, Finlay *et al.* produced an array of squares containing alternating fluorinated and PEGylated stripes of different widths on either a uniform fluorinated or PEGylated background.³⁰ Deposition of fluoro-octadecylchlorosilane (FOTS) was combined with standard lithographic techniques and subsequent back-filling with a PEGylated silane to produce these patterned features. Square arrays measured 1 cm × 1 cm, and stripes with widths of 500, 200, 100, 50, 20, 5, and 2 μm were examined. *Ulva* spores were found to be selective in choosing where to settle, settling at higher densities on fluorinated stripes compared to PEGylated stripes. However, the magnitude of the response was found to be highly dependent on both the width of the stripes and the chemistry of the background, with settlement on fluorinated stripes narrower than 20 μm similar to that experienced for the PEGylated background. This suggests that at some critical dimension below 20 μm, the *Ulva* spores were no longer able to distinguish the difference between the fluorinated and PEGylated features, rather regarding this surface as pure PEG.

Zwitterionic polymers

Inspired by the anti-biofouling properties of blood cell membranes, polymers incorporating zwitterionic molecules such as phosphatidylcholines,^{31,32} are promising as anti-biofouling surfaces. Kitano *et al.* used controlled radical polymerization to polymerize methacrylate monomers with different pendent zwitterionic groups using disulfide carrying *N,N*-diethyldithiocarbamoyl derivatives as chain transfer agents.³³ The resulting oligomers (Fig. 9) formed self-assembled monolayers and brushes on gold surfaces that were found to resist non-specific adsorption of proteins. Using Raman

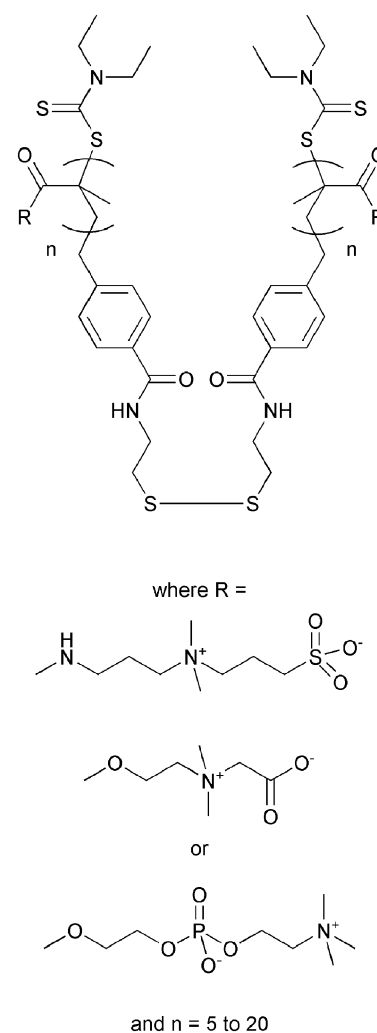


Fig. 9 Antifouling polymers with zwitterionic side chains and a disulfide group for attachment to gold substrates.

spectroscopy and attenuated total reflection infrared spectroscopy, it was determined that the surfaces of zwitterionic polymers did not perturb the native hydrogen bonded network of water in the vicinity of the surface.³⁴ In contrast, the ionic groups and counterions of polyelectrolytes such as poly(sodium acrylate) and poly(sodium ethylenesulfonate) strongly perturbed the structure of water in their hydration shells. This led to the proposal that the existence of the native hydrogen bonded network of water near the surface is a necessary feature for antifouling and biocompatible properties. The exclusion of colloidal and molecular solutes from the vicinity of hydrophilic surfaces, because of the long-range ordering and liquid crystalline state of interfacial water, has been known for

more than 50 years, and has been reviewed by Zheng *et al.*³⁵

Jiang and coworkers prepared polymer brushes of sulfobetaine methacrylate and carboxybeteine methacrylate monomers on glass slides using atom transfer radical polymerization and found that the zwitterionic polymer brushes reduced fibrinogen adsorption to a level comparable to that on PEGylated surfaces.³⁶ These surfaces also resisted adhesion of bovine aortic endothelial cells³⁶ and prevented biofilm formation of Gram positive and Gram negative bacteria.³⁷

As regards the antifouling properties of 'non-zwitterionic' polymers, Chen and Jiang have found in a recent study that charge-balanced polyelectrolytes such as copolymer hydrogels of positively charged aminoethyl methacrylate hydrochloride and negatively charged 2-carboxyethyl acrylate exhibited good resistance to protein adsorption, on par with oligoethylene glycol based coatings.³⁸ They have proposed that the close proximity of amino and carboxylic acid groups in the copolymer makes it similar to a zwitterion in ionic character, and that strong hydration of the copolymer through ionic solvation was key to their nonfouling properties.

Polymers with oligosaccharide grafts

Surfaces coated with oligosaccharide grafted polymers mimic the antifouling glycocalyx, which is an extracellular polysaccharide matrix that prevents non-specific interactions of proteins and cells in the serum with the cell surface.³⁹ Zhu and Marchant have designed and synthesized maltose dendrons and attached these to a poly(vinylamine) backbone to create a canopy of non-adhesive glycocalyx-like saccharide coating over a biomaterial surface (Fig. 10).⁴⁰ The dendritic architecture enabled a dense surface coverage by glycosylated molecules, eliminated potential defect areas, and successfully suppressed platelet adhesion.

Xerogel coatings

Xerogels are glasslike crosslinked networks obtained by the sol-gel process. Tang *et al.* investigated cell settlement and adhesion properties of xerogels with tailored wettability. These xerogels were

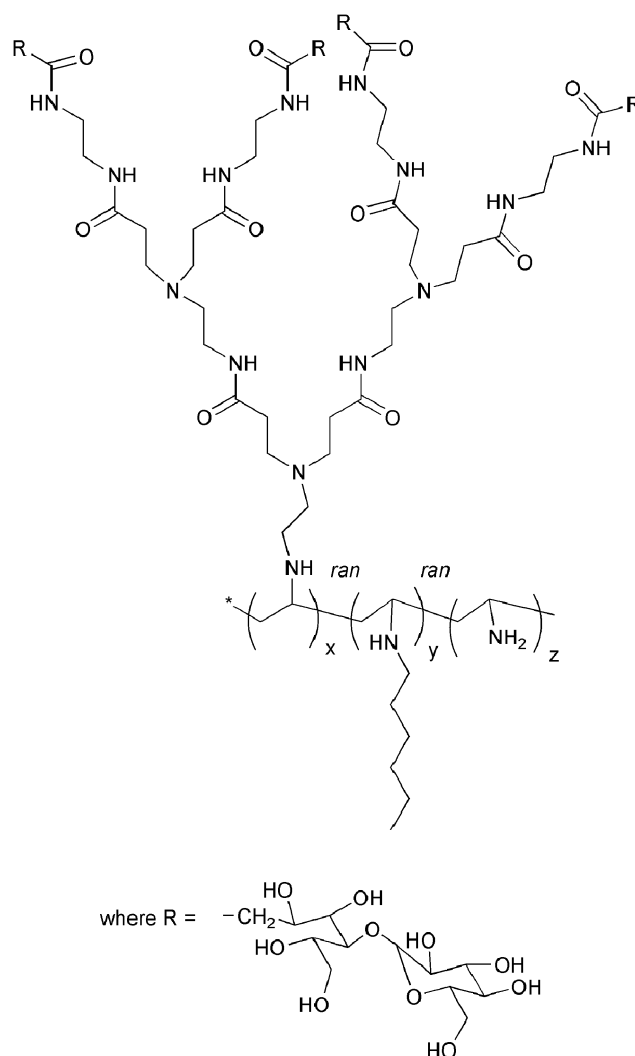


Fig. 10 Antifouling copolymer with glycodendron grafts.

prepared using alkoxy silane precursors containing the non-polar *n*-propyl, *n*-octyl and 3,3,3-trifluoropropyl groups, and the polar ethylenediamine groups (Fig. 11).⁴¹ Coatings incorporating the *n*-octyl groups showed the highest

removal of the eight-day old sporelings of *Ulva*. Coatings with the semifluorinated *n*-propyl groups showed a higher settlement of both algal zoospores and barnacle cyprid larvae. Interestingly, the fluorinated coatings showed only

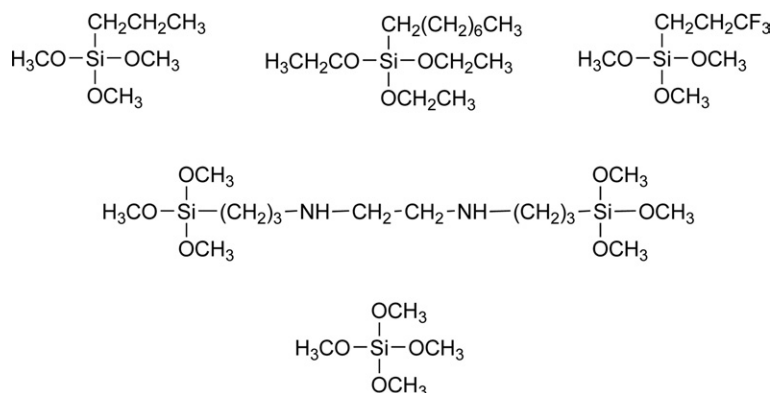


Fig. 11 Precursors for sol-gel synthesis of antifouling coatings.

a moderate release of *Ulva* spores and sporelings, lower than or similar to xerogels containing *n*-propyl and *n*-octyl groups. In comparison with the good release of green algae from liquid crystalline polymers with semifluorinated side chains,^{10,12} the data from fluorinated xerogel coatings suggest that liquid crystallinity and self-assembly of relatively long semifluorinated alkyl side chains⁴² play an important role in fouling release performance.

Siloxane–polyurethane and siloxane–acrylic–polyurethane polymers

To address issues related to poor mechanical properties and poor substrate adhesion of PDMS coatings, siloxane–polyurethane block copolymers have been evaluated for their fouling release properties in marine biofouling assays.^{43,44} Ekin *et al.* have synthesized block copolymers consisting of PDMS and poly(ϵ -caprolactone) of the architectures shown in Fig. 12,⁴³ and formulated polyurethane coatings using a trifunctional alcohol (polycaprolactone triol) and an isocyanurate, whose idealized structures are also shown in the figure. Using a high-throughput synthetic

approach, they evaluated a series of coating formulations containing different molecular weights of the PDMS block and the degrees of polymerization of the ϵ -caprolactone monomer for fouling release properties. The combinatorial approach enabled them to identify coating formulations that showed better fouling release than PDMS controls (Silastic T2 and DC 3140). Coatings prepared using hydroxyalkyl carbamate-terminated PDMS (the linear precursors) showed lower adhesion of barnacles than those prepared using the dihydroxyalkyl carbamate-terminated PDMS (the branched precursors). Lower molecular weight PDMS showed higher removal of soft fouling (bacterial and algal biofilms) than higher molecular weight PDMS. The opposite trend was seen in barnacle adhesion assays. A similar study was conducted wherein a terpolymer of *n*-butyl acrylate, *n*-butyl methacrylate and 2-hydroxyethyl acrylate was used as the polyol.⁴⁴ In addition, 3-aminopropyl-terminated PDMS, and an aliphatic polyisocyanate were used to prepare the coatings. The compositions of the three monomers in the polyol precursor were systematically varied, and siloxane–acrylic–polyurethane formulations resulting in better release of *Navicula* diatoms than control PDMS surfaces were identi-

fied. It was found that increasing the *n*-butylacrylate and 2-hydroxyethylacrylate composition in the acrylic polyol improved fouling release of both cells of the diatom *Navicula* and sporelings of the green macroalga *Ulva*. Some of the experimental observations cannot be fully explained at the present, but it is expected that such a combinatorial approach will eventually help in developing structure–property correlations for biofouling.

Phosphazene polymers

Polyphosphazenes consist of an inorganic –P=N– backbone with organic groups linked to the phosphorous atoms. Poly[bis(2,2,2-trifluoroethoxy)phosphazene] (PTFEP, Fig. 13) is known to be non-thrombogenic in nature.^{45,46} The hydrophobic fluorinated phosphazene polymer,

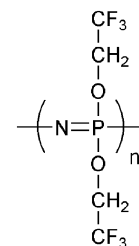


Fig. 13 Poly[bis(2,2,2-trifluoroethoxy)phosphazene].

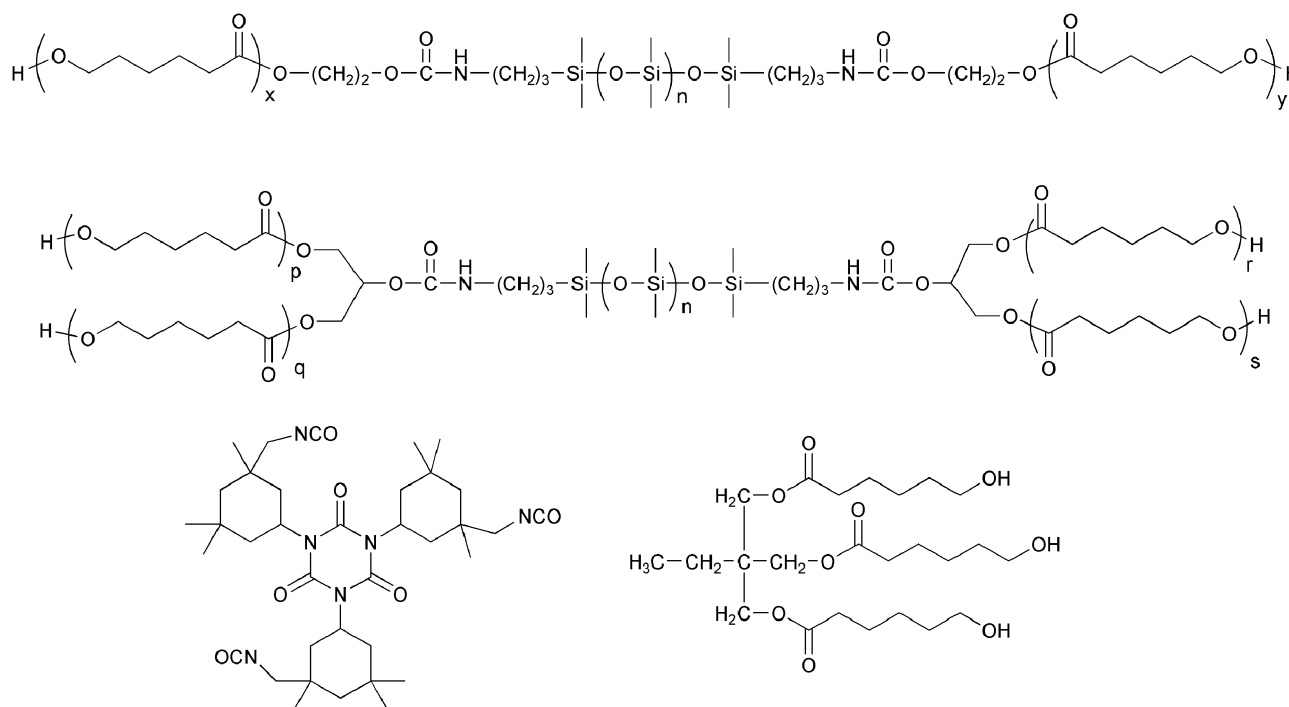


Fig. 12 Formulation ingredients in a siloxane–polyurethane coating system.

with an advancing water contact angle of about 110° , is however not intrinsically antifouling in nature. The resistance to platelet adhesion arises from a 'conditioning film' of human serum albumin, which prevents adsorption of fibrinogen and fibronectin.⁴⁷ In comparison to control surfaces such as poly(methyl methacrylate), silicone, 3-aminopropyltrimethoxy silane SAM, and octyltrichlorosilane SAM, Grunze *et al.* found that the PTFEP surface showed the highest adsorption of human serum albumin and the lowest adsorption of human fibrinogen from a 10% solution of human plasma in phosphate buffer saline.⁴⁷ Thus, the biocompatibility and antifouling property of the hydrophobic polymer is due to a competitive adsorption of albumin that 'passivates' the surface. Long-term antifouling properties of polymeric coatings will strongly depend on the formation of such conditioning layers in the environment they are exposed to.

Polyoxazoline polymers

Polyoxazolines, a well-known class of polymers, have received renewed interest as biomaterials.⁴⁸ These hydrophilic and relatively non-toxic polymers are promising as antifouling coatings for biomedical applications. Konradi *et al.* have investigated the antifouling properties of comb polymers with poly(2-methyl-2-oxazoline) (PMOXA) side chains and a polycationic poly(L-lysine) (PLL) backbone (Fig. 14).²⁷ These polymers were designed in analogy to the PEG-based comb polymers previously investigated by Textor, Spencer, and

coworkers.²¹ The PLL-*g*-PMOXA copolymers spontaneously adsorbed onto metal oxide surfaces, through ionic interactions between the positively charged primary amine groups of the PLL with the negatively charged metal oxide surface. With an optimal side chain grafting density, the resulting adsorbed monolayer of the copolymer could eliminate protein adsorption from full human serum to below 2 ng cm^{-2} , and equalled the protein repellent properties of the best PEG-based coatings. Polymer coatings that are bound to substrates *via* electrostatic interactions are prone to desorption at high salt concentrations,²⁷ and covalent attachment is preferred to overcome this limitation.

Conclusions

We have discussed a wide range of polymeric coatings with promising anti-biofouling and fouling release properties. With an increasing drive toward environmentally friendly non-toxic marine coatings, a large number of non-toxic polymer systems have been designed and evaluated in biofouling assays. Based on the vast set of experimental data that has accumulated so far, certain generalizations can be made regarding the anti-biofouling performance of these coatings. Hydrophilic PEGylated polymers, zwitterionic polymers, and polymers incorporating oligosaccharide moieties are inherently anti-biofouling in nature and resist protein adsorption and cell settlement. Hydrophobic PDMS and fluoropolymer surfaces are not resistant to protein adsorption, but are non-adhesive to cells and organisms because of

their non-polar nature. Amphiphilic polymers incorporating hydrophilic PEGylated groups and hydrophobic fluorinated groups have shown anti-biofouling and fouling release properties, comparable to or better than the widely used PDMS elastomers. Because of their relatively low cost, there is a significant ongoing effort in improving the anti-biofouling properties of PDMS elastomers, especially against diatom fouling. Additionally, increasing the yield strength and toughness of PDMS elastomers is of great importance. Even with the relatively expensive fluorinated polymers, surface segregation of the low-energy fluoroalkyl groups allows blending with an inexpensive polymer, a viable approach for creating anti-biofouling coatings. Zwitterionic polymers are a unique class of charged polymers that show reduced protein adsorption and cell adhesion. Other ionically charged polymers such as those consisting of pyridinium groups⁴⁹ are not resistant to protein adsorption because of electrostatic interactions. In fact, adhesion and close contact of cells with surfaces of pyridinium polymers, and other polycations,⁵⁰ are believed to be important for their antimicrobial activity. The future direction of anti-biofouling research should be toward correlating molecular level details of a surface with anti-biofouling behavior, and establishing a fundamental understanding of anti-biofouling and fouling release mechanisms. Understanding why certain organisms such as diatoms adhere strongly to non-polar PDMS coatings (which are otherwise non-adhesive in nature) will require knowledge on the role of mechanical properties of the coating on fouling release and the chemical composition of the adhesive matrices secreted by these organisms.

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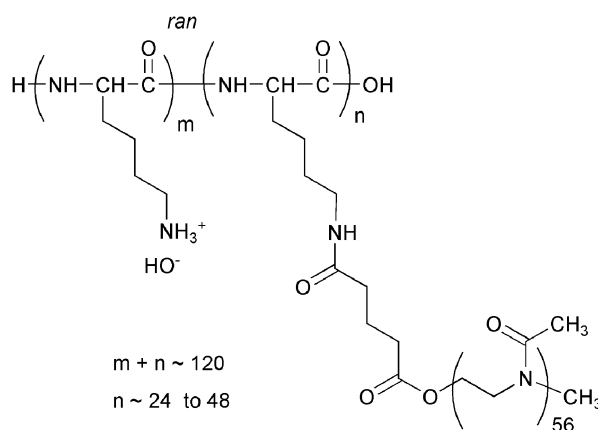


Fig. 14 Poly(L-lysine)-graft-poly(2-methyl-2-oxazoline).

Barbara, James and Maureen Callow at the University of Birmingham, and Dean Wendt at California Polytechnic State University.

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